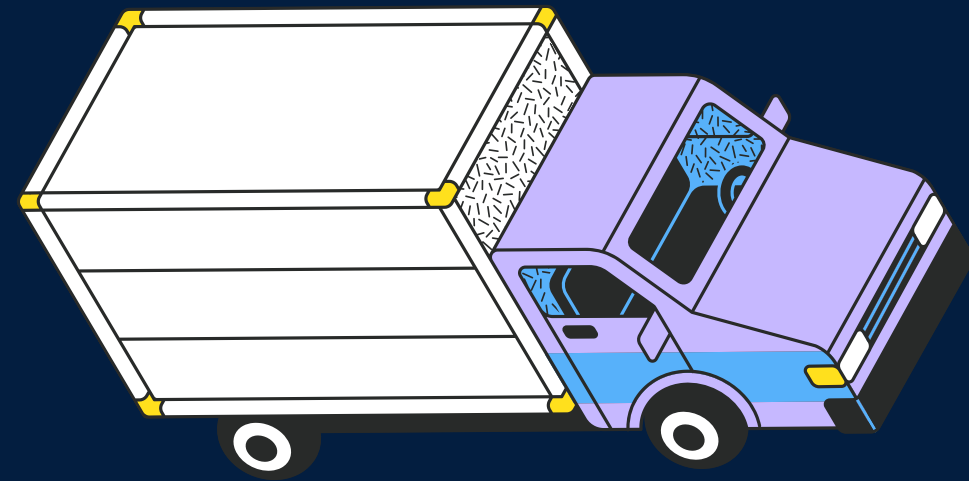


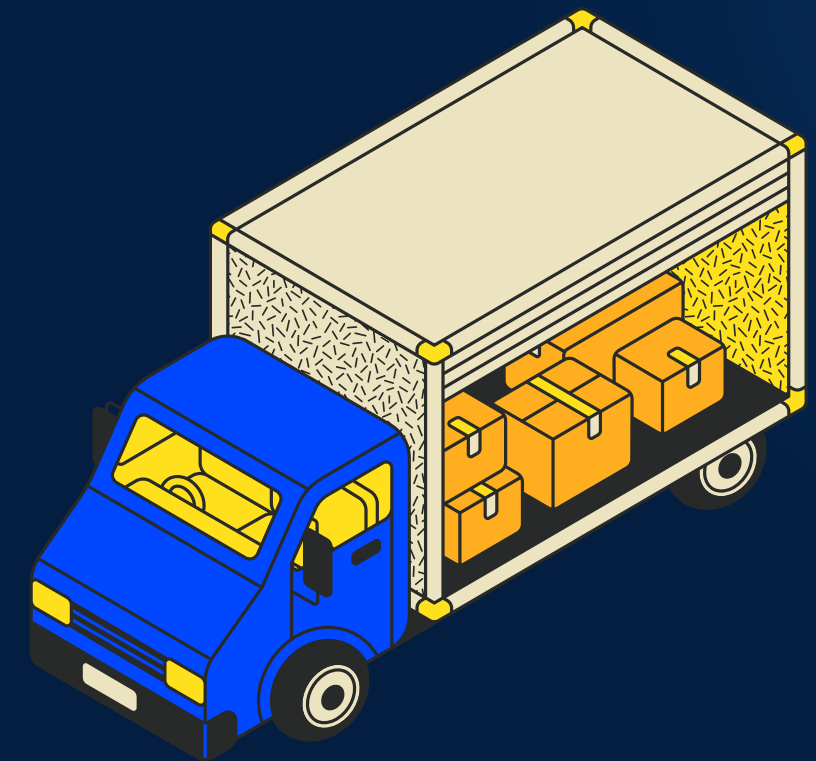


LOGISTICS AND DELIVERY PERFORMANCE ANALYSIS

ETL

Presented by:

Ahmed Essam eldin
Abdulrahman Hamada
Toka Hany



OVERVIEW

01

Data Model

02

Transformation

03

Dashboard

01 Data Model



Tables

- System consist of 5 tables:
- Customers → customer's information.
- Products → product details
- Orders → contain order's data
- Logistics → contain orders delivery dates
- Staggering-OrderLines → considered as fact table that record sales and transactions.



Data Generation

- Generation of data using AI.
- Filling 100 rows logistics and orderLines tables.
- Allowing Null values in estimated_delivery_date & Actual_delivery_date.
- Make sure that not both columns contain Null in a row.
- Actual_delivery_date must be equal or more than estimated_delivery_date.

1.2 STAR SCHEMA

01

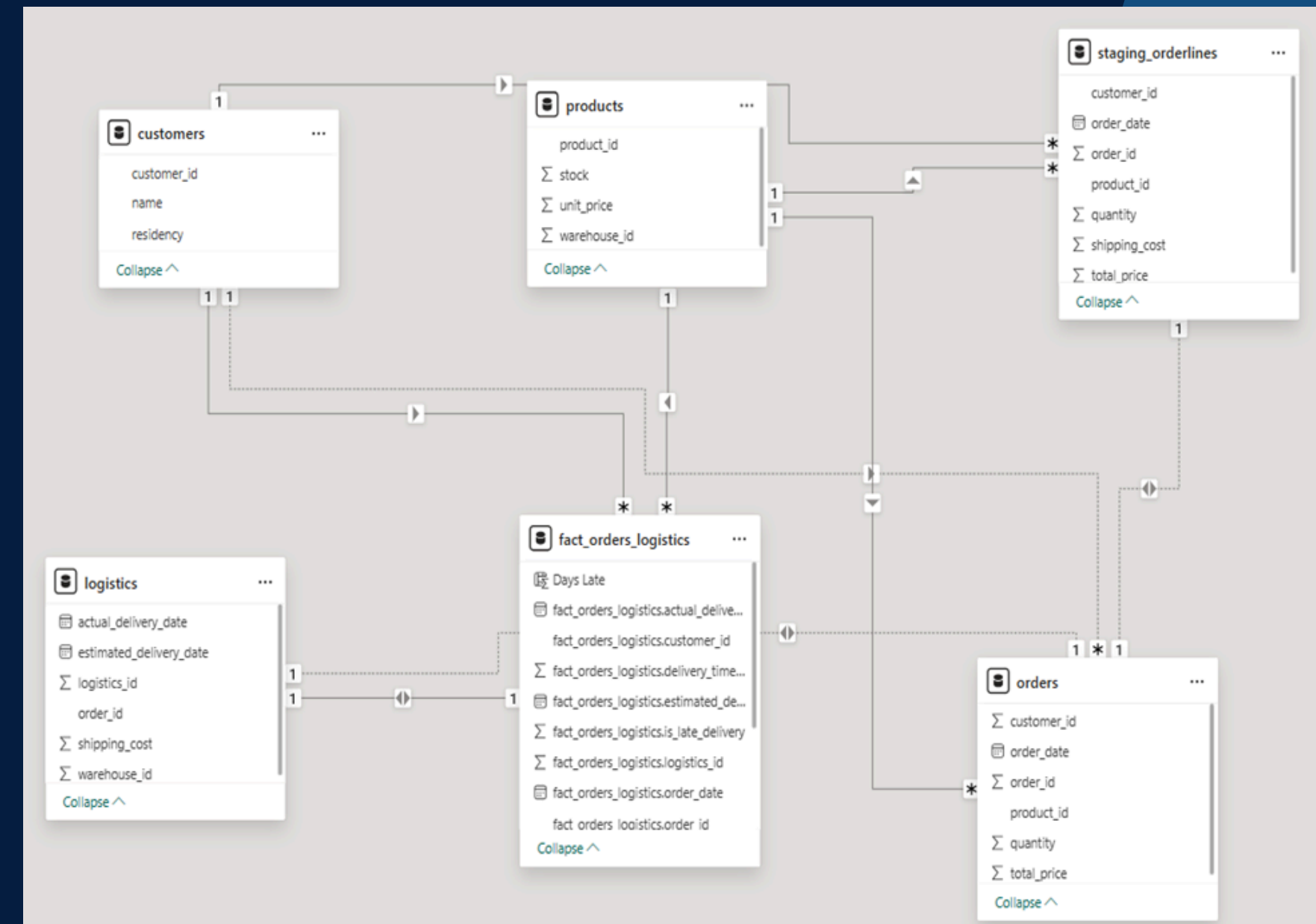
Fact_orders_Logistics considered as the fact table that results from joining orders and logistic table

02

Stagging_OrderLines considered as Dimension Table that contain Customer_id, Order_id, Product_id and Order_date

03

Logistics Dimension Table contains warehouse_id, actual & estimated delivery_date,



1.3

DATABASE

Creating Tables in PostgreSQL

Customer Table

```
CREATE TABLE customer (  
  customer_id SERIAL PRIMARY  
  KEY,  
  name VARCHAR(100) NOT NULL,  
  residency VARCHAR(100)  
);
```

Inserting Values

Customers

```
COPY customer(customer_id, name, residency)  
FROM '/var/lib/postgresql/data/customers.csv'  
DELIMITER ',' CSV HEADER;
```

02 TRANSFORMATION

The Goal & Core KPIs

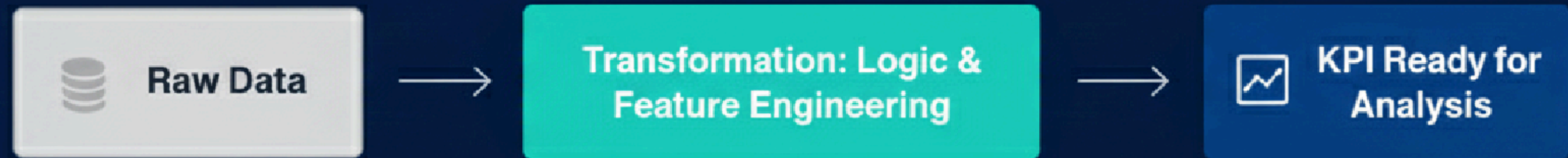
The main objective of our transformation phase was critical: not just cleaning data, but deriving **meaningful metrics** directly from the **raw inputs**. We focused on creating features that directly address our business question: How efficient is our delivery process?

Customer Satisfaction

The central KPI for this entire project is the **Delivery Time Delta**—the difference between the `actual_delivery_date` and the `estimated_delivery_date`.



2.1 Data Transformation: Building the core KPI



Core KPI
- Delivery Time Delta

```
datediff(col"actual_delivery_date",  
         estimated_delivery_date))
```

2.2 Feature Engineering & Data Integration

From Delta to Flag: Defining Delivery Success

Engineered Feature

- is_late delivery flag
(1=Late, 0 On time)

```
when(col("actual_date") > col  
estimated_deate), 1.  
1).othowive(0)
```

Data Integration



Integrated Fact Table:
`staging.fact\_orders\_logistics` (with
Dimension Keys & Metrics)"

2.3 Technical Output & Staging Zone

PySpark ETL Pipeline: Persisting Data for Analysis

Technical Implementation



PySpark ETL Script

- Data Quality:
`isNull()` checks for dates

Staging Zone Output



Data persisted in `staging` database
(Main Fact Table)

staging.fact_orders_logistics
staging.logistics_clean
staging.customer_clean



3

DASHBOARD



OVERVIEW

DASHBOARD SHOWS ANALYSIS OF SALES, ORDERS, SHIPPING COSTS, AND DELIVERY PERFORMANCE

OVERVIEW & KPIS



KPIS

- ▶ Orders: 100
- ▶ Total Sales: 107.59K
- ▶ Avg. Shipping Cost: 28.87
- ▶ Late Orders: 60%
- ▶ Avg. Days Late: 1.64
- ▶ Geography Insight: Majority of orders came from North America and Europe

Num.of orders

100

Total Sales

107.59K

Avg_shipping cost

28.87

late_orders

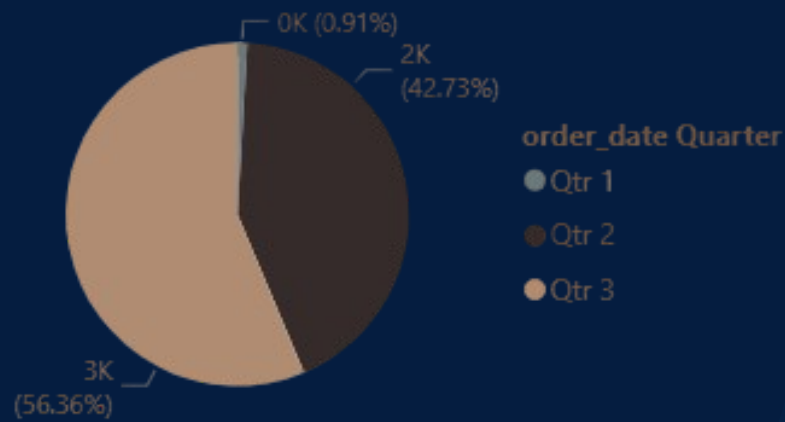
60

Avg_Days late

1.64

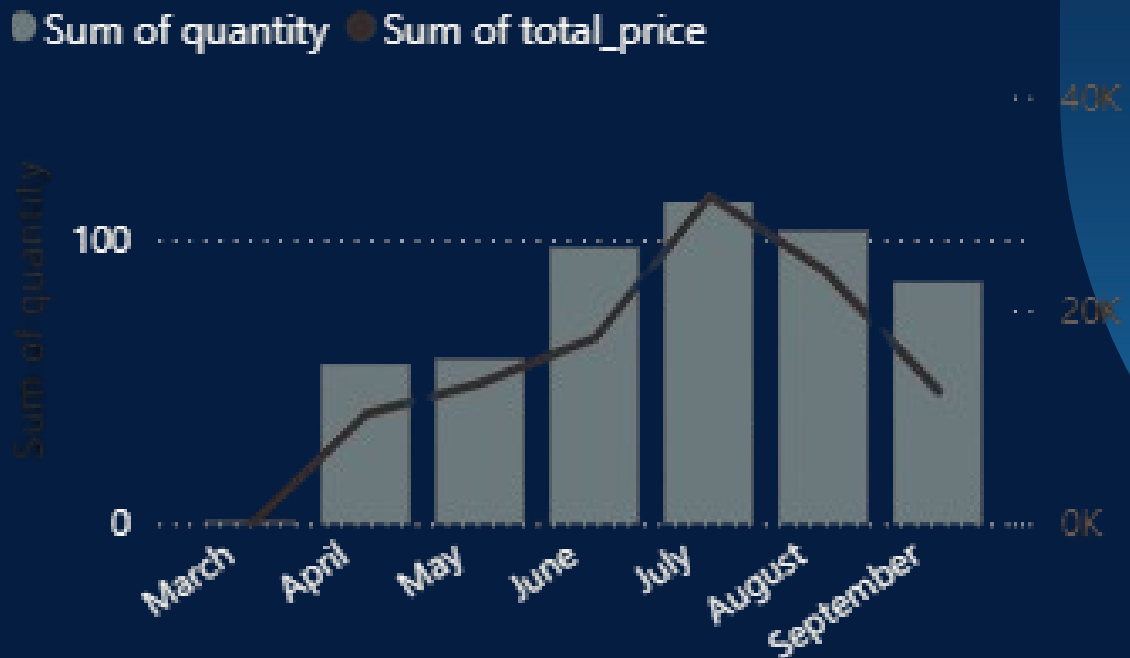
SALES & ORDERS ANALYSIS

Quarterly Trend: Most sales and orders occurred in Q2 and Q3



On a monthly level, July and August recorded the highest sales and quantities, confirming that summer is the peak season for business activity

Although summer showed strong performance, September experienced the highest delivery delays and the lowest profit compared to July and August, which may indicate that late deliveries directly impacted customer satisfaction and sales.



RECOMMENDATIONS

I

Improve shipping and supply chain management to reduce delays

II

Enhance customer experience to maintain loyalty despite delivery issues

III

Leverage Q2 and Q3 with more promotions and marketing campaigns

IV

Geographic Strategy: Strengthen our presence in strong markets like North America and Europe through targeted marketing and offers, while also investing in underperforming regions to unlock new growth opportunities.

THANK YOU

